Forward's vacuum-fluctuation battery

Can charged leaves pulled together by the Casimir force deliver electrical energy, and is it a continuous source? — register entry H25

kronspice simulation report · 2026-11-07 · 100 nm Al leaves, 1 μm → 20 nm at 1 mm/s

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Summary

R. L. Forward (*Phys. Rev. B* 30, 1700, 1984) proposed a battery driven by the Casimir force, the attraction between close metal plates that comes from the electromagnetic zero-point fluctuations of empty space. A stack of thin aluminium leaves carries equal charges, so their electric repulsion pushes them apart; a fast feedback (servo) supply holds that repulsion just below the Casimir pull, so the leaves creep together, and the work the Casimir force does goes into the supply as electrical energy. Pushing the leaves apart again “recharges” it. This example computes the pull for real aluminium, models one leaf pair as a Kron (q, a) system with a full ledger, twins its electrical side in kronspice, and runs a full cycle.

- **Real aluminium pulls less than perfect mirrors:** at 20 nm the Lifshitz energy is 0.230 of Forward's ideal value (1.245×10^{-5} against 5.417×10^{-5} J/m²).
- **One closing stroke gains energy, once:** 12.3 μJ/m² for aluminium (54.0 μJ/m² for perfect mirrors), equal to the Casimir work less servo losses.
- **The supply pays first:** during the stroke it puts in 4.22 mJ/m² (hundreds of times the Casimir work), stored in the field; the gain appears only when the leaves are stopped and discharged.
- **A full cycle is a net loss:** reopening costs 12.6 μJ/m², so close + reopen gives −0.28 μJ/m² (aluminium). The Casimir force is conservative.
- **kronspice twin agrees:** supply energy 4.21981 mJ against 4.22084 mJ from the model's own solution; the leaves' work on the capacitor equals the Casimir work (12.4543 against 12.4542 μJ).
- **Scale:** 0.023 J per kg of leaf metal per stroke, against about 900,000 J/kg for a lithium-ion cell.

What is being tested

Hypothesis (H25). Electrical energy can be extracted from the vacuum by the Casimir cohesion of charged foliated conductors (Forward 1984). Forward himself compared it to a hydroelectric dam: the force is a catalyst that releases stored energy, here the surface energy of the separated leaves.

Standard prediction. The Casimir force is conservative, so one stroke releases stored surface energy once, consistent with energy conservation (Cole & Puthoff, *Phys. Rev. E* 48, 1562, 1993); reopening the leaves costs at least what closing gave. The energy cannot exceed the leaves' surface energy, about 1.1 J/m² per aluminium surface (Tyson & Miller 1977).

1. How strong is the pull?

Forward used the formulas for perfect mirrors (his eqs. 1–2). Real aluminium stops reflecting above its plasma frequency, so the Lifshitz formula is used with aluminium's plasma wavelength λ_p = 107 nm (Lambrecht & Reynaud 2000). With λ_p set to zero the calculation reproduces Forward's eq. 1 to 10^{-5}. Energies and pressures are per m² of plate; negative values mean attraction.

gap	energy, ideal	energy, Al	Al ÷ ideal	pressure, ideal	pressure, Al
10 nm	-4.334×10^{-4} J	-5.861×10^{-5} J	0.135	-1.300×10^{5} Pa	-1.284×10^{4} Pa
20 nm	-5.417×10^{-5} J	-1.245×10^{-5} J	0.230	-8.126×10^{3} Pa	-1.421×10^{3} Pa
100 nm	-4.334×10^{-7} J	-2.547×10^{-7} J	0.588	-1.300×10^{1} Pa	−6.586 Pa
1000 nm	-4.334×10^{-10} J	-4.056×10^{-10} J	0.936	-1.300×10^{-3} Pa	-1.191×10^{-3} Pa

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Below λ/2π the force turns to the van der Waals form (Forward eq. 4); the plasma model includes it.

2. One leaf pair as a Kron circuit

Kron's coordinates are the charge q on the leaves and the gap a, as in his machines (Hoffmann 1949; see *hoffmann_motor*). Both leaves sit at the supply voltage V, so there is no field in the gap; each faces a grounded neighbour at distance L, giving $C(a) = 4\varepsilon_0/(L - a)$ per m². The field outside pushes the leaves apart: Forward's repulsion. The equations are Lagrange–Maxwell with Casimir's energy as a potential (r: supply resistance, m: leaf mass per m²):

```
r·dq/dt + q/C(a) = V
½m·d²a/dt² = (electric push) − (Casimir pull)
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The servo sets V at every instant so that the push equals the pull, plus just enough to move the leaves at a steady 1 mm/s. Declared values: 100 nm aluminium leaves, 2 μm to the grounded neighbours, closing from 1 μm to 20 nm.

per m² of gap	perfect mirrors	aluminium
work done by the Casimir force	5.4173×10^{-5} J	1.2454×10^{-5} J
servo voltage at 20 nm	63.8 V	26.7 V
energy the supply gives during the stroke	2.4162×10^{-2} J	4.2208×10^{-3} J
energy stored in the field at 20 nm	2.4216×10^{-2} J	4.2332×10^{-3} J
net gain after discharging at the stop	**5.4023×10^{-5} J**	**1.2315×10^{-5} J**
losses: heat; motion; ledger residual	2.9×10^{-8}; 1.5×10^{-10}; 1.2×10^{-7} J	2.4×10^{-8}; 7.8×10^{-11}; 1.2×10^{-7} J
charge sent during the stroke	7.587×10^{-4} C	3.172×10^{-4} C
electrons returned at discharge	4.74×10^{15}	1.98×10^{15}
Casimir energy carried out per electron	0.071 eV	0.039 eV

- **During the stroke the supply pays.** The Casimir pull grows as $1/a^4$, so the servo must raise the voltage steeply; the supply puts in hundreds of times the Casimir work, and it is stored in the field.
- **The gain appears only at the end.** When the leaves are stopped and discharged, all the charge comes back (net charge moved over the whole operation: zero), each electron carrying a few hundredths of an eV more than it went in with. Net, the supply gains the Casimir work less tiny servo losses.
- **The ledger closes** to about 1 % of the Casimir work and 3×10^{-5} of the energy flows (residual 1.2×10^{-7} J/m²).

3. The same electrical side in kronspice

kronspice's varying capacitor $C(t) = 4\varepsilon_0/(L - a(t))$ follows the gap of the stroke and is driven by the servo voltage through the supply resistance (1 μΩ·m²).

	kronspice	own solution	ledger mismatch
perfect mirrors: supply energy	2.41528×10^{-2} J	2.41615×10^{-2} J	4×10^{-4}
perfect mirrors: charge sent	7.58463×10^{-4} C	7.58679×10^{-4} C	
perfect mirrors: leaves' work on the capacitor	5.41731×10^{-5} J	Casimir 5.41731×10^{-5} J	
aluminium: supply energy	4.21981×10^{-3} J	4.22084×10^{-3} J	2×10^{-4}
aluminium: charge sent	3.17151×10^{-4} C	3.17207×10^{-4} C	
aluminium: leaves' work on the capacitor	1.24543×10^{-5} J	Casimir 1.24542×10^{-5} J	

4. A full cycle: close, then reopen

per m²	closing gain	reopening cost	cycle
perfect mirrors	$+5.4023\times10^{-5}$ J	-5.4311×10^{-5} J	**-2.884×10^{-7} J**
aluminium	$+1.2315\times10^{-5}$ J	-1.2592×10^{-5} J	**-2.771×10^{-7} J**

Reopening costs what closing gave, and the servo's losses make every cycle a small net loss. Forward's battery is a one-shot release of stored surface energy, like a spring — not a continuous source.

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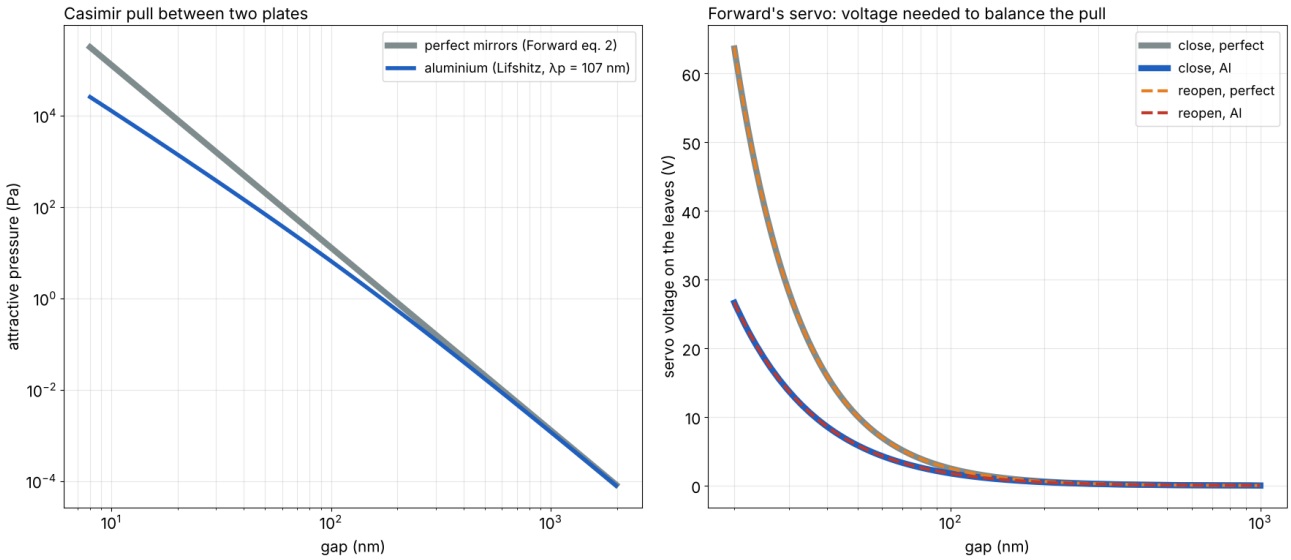


Figure 1. Left: attractive Casimir pressure against gap for perfect mirrors (Forward eq. 2, grey) and aluminium (Lifshitz, λ_p = 107 nm, blue); real metal pulls less at small gaps. Right: the servo voltage needed to balance the pull, closing (solid) and reopening (dashed), for perfect mirrors and aluminium.

5. Scale, and electron pumps

- **Energy per kilogram.** One stroke of real aluminium yields 1.23×10^{-5} J per m² of gap, 0.023 J per kg of leaf metal (100 nm leaves, two per gap). A lithium-ion cell stores about 900,000 J/kg.
- **The upper limit.** Pressing the leaves into full contact cannot release more than their surface energy, about 1.1 J/m² per aluminium surface (Tyson & Miller 1977) — the same energy that had to be spent to make the separate leaves.
- **As an electron pump.** One stroke is a variable-capacitor charge pump driven by the Casimir force: it lifts 2.0×10^{15} electrons per m² by 0.039 eV each, once; reopening costs that back. A continuous pump needs a continuous source of work, as the pumps of *photon_electron* have.
- **What it is good for.** The servo-controlled variable capacitor itself, and the Casimir force as a nanoscale actuator. It is not a source of energy.
- **Measurements.** The Casimir force between parallel metal plates has been measured (Bressi et al., *PRL* 88, 041804, 2002, 0.5–3 μm, 15 %). Forward noted that no measurements existed for metal plates at the 1–20 nm spacings his battery needs.

Validation and limits

main_test.go checks the perfect-mirror limit (Forward eq. 1 to 10^{-5}), the stroke ledger and its kronspice twin, and the cycle. Scope: one leaf pair; aluminium in the plasma model (λ_p = 107 nm); a servo moving the leaves at a steady 1 mm/s; declared leaf thickness and spacing.

Status

H25: a one-shot release of surface (Casimir) energy, consistent with energy conservation (Cole & Puthoff 1993); 12 μJ/m² per aluminium stroke, 0.02 J/kg; a close–reopen cycle is a net loss (computed) — not a continuous source. The ledger closes: what the supply gains is the leaves' surface energy.

Reproducing the results

```
go run ./examples/forward_casimir -viz examples/forward_casimir/viz   # about 3 seconds
python3 examples/forward_casimir/viz/render.py
go test ./examples/forward_casimir
python examples/forward_casimir/viz/make_report.py examples/forward_casimir/forward_casimir_report.pdf
```

Sources: R. L. Forward, Phys. Rev. B 30, 1700 (1984); H. B. G. Casimir, Proc. K. Ned. Akad. Wet. 51, 793 (1948); E. M. Lifshitz, Sov. Phys. JETP 2, 73 (1956); A. Lambrecht, S. Reynaud, Eur. Phys. J. D 8, 309 (2000); D. C. Cole, H. E. Puthoff, Phys. Rev. E 48, 1562 (1993); G. Bressi et al., PRL 88, 041804 (2002); W. R. Tyson, W. A. Miller, Surface Science 62, 267 (1977).

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